SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO4: *Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)*

Assessment Tool Weightage: 50%

Duration : 1 hour

QUESTIONS: 4 **Total Marks:40**

Annexure A

ARTICLE REVIEW

Code of Conduct:

I **LATISHA S – 192524071** *Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

ARTICLE REVIEW

INSTRUCTIONS

Select any ONE peer-reviewed research article (published after 2019) from IEEE Xplore, Springer, Elsevier, or ACM Digital Library related to any of the following topics: *Inductive Learning, Decision Trees, Statistical Learning Methods (Naïve Bayes / Logistic Regression / SVM), or Reinforcement Learning applications.*

Submit: (i) Article citation details (ii) PDF or valid DOI link (iii) Written review as per the tasks below.

Task 1: Article Summary

(a) Provide the full citation of the article (Author(s), Title, Journal/Conference, Year, DOI). State the domain/application area and the specific ML problem addressed.

(b) Summarise the key objective of the article in your own words (150–200 words). Clearly state the research gap the authors identified and how they proposed to fill it.

Task 2: Methodology Analysis

(a) Explain the ML technique(s) used in the article (e.g., Decision Tree variant, RL algorithm, Statistical model). Describe the dataset used—size, features, source, and how it was prepared.

(b) Map the technique to CO 4 topics (Inductive Learning / Decision Trees / Statistical Learning / RL). Identify one methodological strength and one limitation in the authors' approach.

Task 3: Results & Critical Evaluation

(a) Report the key results (accuracy, F1-Score, AUC-ROC, reward convergence, etc.) as presented by the authors. Create a comparative table if multiple models are benchmarked.

(b) Critically evaluate the results: Are the reported metrics realistic? Are there potential biases (dataset, evaluation protocol)? What additional experiments would strengthen the findings?

(c) Discuss real-world applicability: Where can this technique be deployed? What are the challenges in scaling the solution to industry (data availability, compute, ethical issues)?

Task 4: Reflection & Learning Outcome

(a) State three key insights you gained from reading the article that deepen your understanding of CO 4 topics. Connect each insight to a specific concept from the course syllabus.

(b) If you were to extend this research, propose one novel experiment or enhancement (different dataset / improved algorithm / new application domain). Justify its feasibility and expected impact.

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Article	20%	Demonstrates clear and in-depth understanding of the article's purpose, concepts, and arguments	Shows good understanding with minor gaps	Partial understanding; misses key ideas	Lacks understanding of the article
Summary	20%	Provides concise and accurate summary highlighting all key points	Covers most key points with minor omissions	Incomplete or partially accurate summary	Poor or incorrect summary
Critical Analysis	25%	Provides insightful critique with strong reasoning and evaluation	Provides reasonable analysis with some justification	Superficial analysis; limited evaluation	No meaningful analysis
Use of Evidence	15%	Effectively uses examples, data, or references to support review	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Organization	20%	Well-structured, clear, and coherent writing with proper language and flow	Generally clear with minor issues	Poor organization; lacks clarity	Disorganized and difficult to understand

Task 1: Article Summary

(a) Domain and ML Problem

The article focuses on healthcare, specifically the prediction of Type-2 diabetes. The authors studied different demographic, lifestyle, hereditary, and medical risk factors and evaluated machine-learning algorithms for classifying patients as diabetic or non-diabetic.

The main ML problem is:

Given patient risk factors, predict whether the patient is likely to have Type-2 diabetes.

(b) Key Objective

The main objective of the article is to provide a fair comparison of different machine-learning algorithms for Type-2 diabetes prediction. The authors identified a research gap in previous studies: different researchers used different datasets, preprocessing methods, and evaluation measures, making it difficult to determine which algorithms actually perform better. To address this gap, the authors created a unified experimental setup and evaluated 35 machine-learning algorithms using three real-world diabetes datasets. They also investigated whether feature-selection techniques could improve prediction performance and reduce computational time. The study considered risk factors from demographic, hereditary, lifestyle, and medical categories. The algorithms were evaluated mainly using accuracy, F-measure, and execution time. The study found that Bagging with Logistic Regression performed best on the PIMA Indian dataset, while Random Forest performed best on the highly imbalanced UCI and MIMIC-III datasets. The authors also showed that accuracy alone can be misleading for imbalanced medical datasets and recommended using F-measure along with accuracy.

Task 2: Methodology Analysis

(a) ML Techniques and Dataset

The study evaluated 35 algorithms, including:

- Decision Tree
- Naïve Bayes
- Logistic Regression
- SVM
- KNN
- Random Forest
- MLP
- Bagging
- Boosting
- Stacking
- K-Means + classification

Decision Tree selects features based on Information Gain, while Logistic Regression predicts diabetes probability using a sigmoid function.

Datasets

Dataset	Description
PIMA Indian Diabetes	Diabetes diagnostic measurements from the National Institute of Diabetes and Digestive and Kidney Diseases
UCI	Patient/hospital data from 130 US hospitals
MIMIC-III	Data from more than 40,000 ICU patients

The authors used Weka 3.8, feature-selection methods, and 10-fold cross-validation. Each algorithm was executed three times and average performance was reported.

(b) Mapping to CO4

CO4 Topic	Connection
Inductive Learning	Models learn patterns from existing patient records to predict new cases.
Decision Trees	Decision Tree and Random Forest were evaluated.
Statistical Learning	Logistic Regression and Naïve Bayes were included.
Feature Selection	Important diabetes risk factors were identified.

Methodological Strength

Strong point: The authors used three different datasets and a common evaluation framework, making model comparison more meaningful.

Limitation

The datasets have significant class imbalance, especially UCI and MIMIC-III. Several algorithms failed to predict the minority diabetes class, showing that accuracy alone is insufficient.

Task 3: Results & Critical Evaluation

(a) Key Results

The major findings were:

Dataset	Best-performing approach
PIMA Indian	Bagging + Logistic Regression
UCI	Random Forest
MIMIC-III	Random Forest

The authors reported that Random Forest performed particularly well on imbalanced datasets, while Bagging-LR performed best for the PIMA dataset.

Feature selection also reduced execution time while maintaining prediction performance.

Important Finding

The paper specifically warns that high accuracy does not necessarily mean good medical performance. Some models achieved high accuracy while failing to predict the minority diabetic class. Therefore, F-measure is important for imbalanced healthcare data.

(b) Critical Evaluation

Are the results realistic?

Yes, the results are reasonably realistic because the study uses multiple real-world datasets and repeated cross-validation. However, performance can vary significantly depending on the dataset.

Potential Biases

- Dataset imbalance may favour the majority class.
- PIMA and other datasets may not represent every population.
- Different healthcare systems may have different patient characteristics.
- Historical hospital data may contain demographic or selection bias.

Additional Experiments Suggested

The study could be strengthened by:

1. Testing on a completely independent external hospital dataset.
2. Reporting ROC-AUC, sensitivity and specificity more extensively.
3. Using SMOTE/class-weighting to handle imbalance.
4. Testing the models across different demographic groups.

(c) Real-World Applicability

The approach can be used in:

- Hospital diabetes screening
- Electronic Health Record systems
- Health-risk assessment applications
- Primary healthcare screening
- Remote patient monitoring
- Clinical decision-support systems

Industry Challenges

Data availability: Large, high-quality labelled datasets are required.

Scalability: Large hospital datasets require efficient storage and computing resources.

Privacy: Medical records must be protected.

Bias: Models must work fairly across different populations.

Explainability: Doctors need understandable reasons behind predictions.

Clinical validation: A model should undergo extensive external validation before being used for real medical decisions.

Task 4: Reflection & Learning Outcome

(a) Three Key Insights

1. Accuracy is not always enough

I learned that accuracy can be misleading when the dataset is imbalanced.

CO4 Connection: Statistical Learning / Classification Evaluation.

In healthcare, F1-score, recall and sensitivity can be more meaningful than accuracy alone.

2. Decision Trees use feature importance

The study helped me understand that Decision Trees select features based on Information Gain and use them to form decision rules.

CO4 Connection: Decision Trees.

This makes Decision Trees relatively easy to interpret compared with complex models.

3. Model performance depends on the dataset

The best algorithm was not the same for every dataset. Bagging-LR performed best for PIMA, while Random Forest was better for the imbalanced UCI and MIMIC-III datasets.

CO4 Connection: Inductive Learning / Statistical Learning.

There is no universally best ML algorithm; performance depends on data characteristics.

Task 4(b): Proposed Extension

Proposed Experiment

I would extend the study by applying SMOTE + Explainable AI (SHAP) to the diabetes datasets.

Approach

Patient Dataset

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Data Preprocessing

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SMOTE for Class Balancing

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Random Forest / Logistic Regression

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SHAP Feature Explanation

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Diabetes Risk Prediction

Justification

SMOTE could improve detection of minority diabetic cases, while SHAP could explain which features contributed most to an individual prediction.

Expected Impact

- Higher diabetes recall
- Better handling of class imbalance
- Improved interpretability
- Greater clinician trust
- More practical clinical decision support